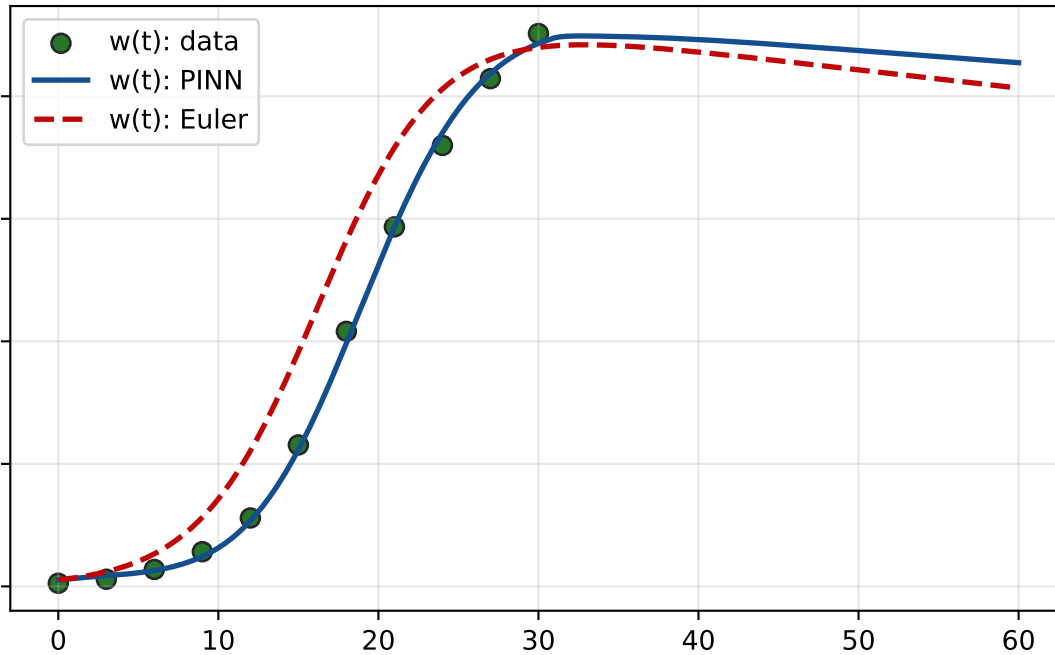
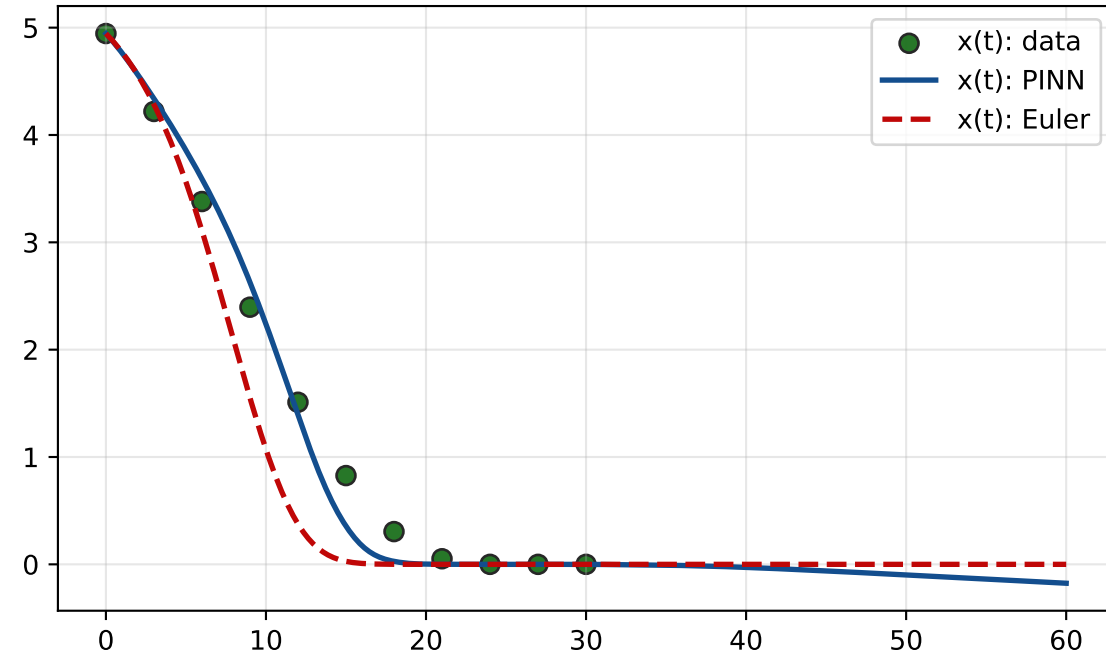


# E8 | PINN vs Euler

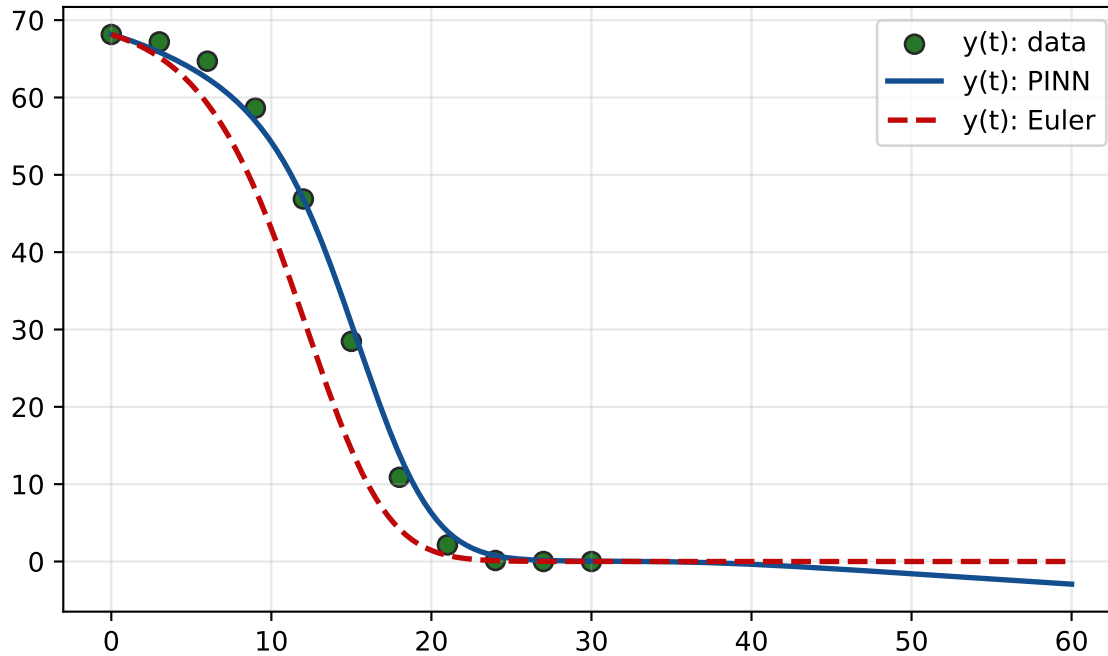
w(t): Biomass



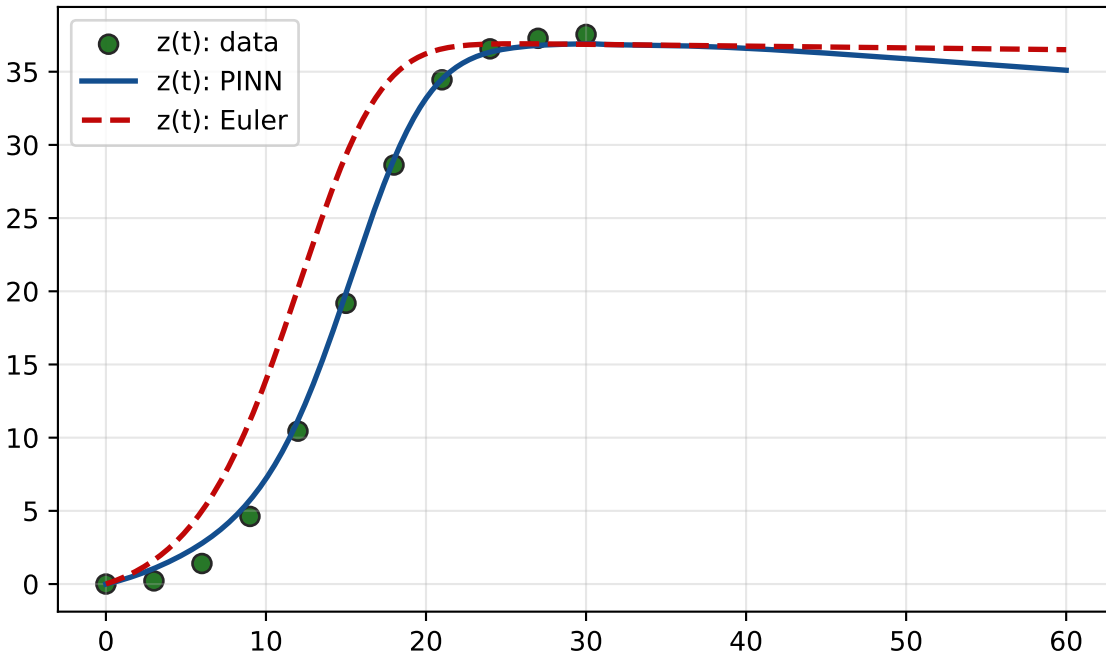
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E8

R2=0.9203, R2\_adj=0.9074, RSS=9.350e+02, AIC=1.488e+02

cond(JTJ)=8.69e+04, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 2.634686e-01 | 3.882411e-01 | 7.859530e-01 | -5.224844e-01 | 1.049422e+00 | 5.014902e-01 |
| p2    | 2.835610e-02 | 3.909254e-02 | 7.913871e-02 | -5.078261e-02 | 1.074948e-01 | 4.726759e-01 |
| p3    | 1.397428e-02 | 5.095228e-01 | 1.031475e+00 | -1.017501e+00 | 1.045449e+00 | 9.782633e-01 |
| p5    | 2.911324e-01 | 5.824542e-01 | 1.179117e+00 | -8.879844e-01 | 1.470249e+00 | 6.200717e-01 |
| p7    | 8.674141e-02 | 6.030773e-02 | 1.220866e-01 | -3.534520e-02 | 2.088280e-01 | 1.585307e-01 |
| p9    | 4.644498e-02 | 3.193426e-02 | 6.464753e-02 | -1.820255e-02 | 1.110925e-01 | 1.540479e-01 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 1.061224e-01 |
| x0 (fixed)     | 4.945277e+00 |
| y0 (fixed)     | 6.814504e+01 |
| z0 (fixed)     | 0.000000e+00 |
| W0 (fixed)     | 0.000000e+00 |